

Diabetes Mellitus MCQ Bank

CBME Assessment - Multiple Choice Questions

Section 1: Epidemiology & Pathophysiology

Question 1:

What is the current global prevalence of diabetes mellitus (2023)?

1. 337 million people (IDF Atlas 9th Edition)
2. 463 million people (IDF Atlas 9th Edition)
3. 537 million people (IDF Atlas 10th Edition)
4. 611 million people (IDF Atlas 10th Edition)

Answer: 3 (*IDF Atlas 10th Edition reports 537 million adults with diabetes globally*)

Question 2:

In India, approximately what percentage of adults aged 20-79 years have diabetes?

1. 5.3%
2. 8.9%
3. 12.1%
4. 15.3%

Answer: 4 (*77 million out of 500 million adults 20-79 years, approximately 15.3%*)

Question 3:

Which of the following best describes the pathophysiology of Type 2 diabetes mellitus?

1. Autoimmune destruction of pancreatic β -cells
2. Insulin resistance combined with relative insulin deficiency
3. Complete absence of insulin secretion due to genetic mutation
4. Pancreatic damage due to chronic pancreatitis

Answer: 2 (*Type 2 DM involves both insulin resistance and progressive β -cell dysfunction*)

Question 4:

According to ADA 2023 guidelines, which HbA1c level confirms diabetes diagnosis?

1. $\geq 6.0\%$ (42 mmol/mol)
2. $\geq 6.5\%$ (48 mmol/mol)
3. $\geq 7.0\%$ (53 mmol/mol)
4. $\geq 7.5\%$ (58 mmol/mol)

Answer: 2 (*ADA defines diabetes as HbA1c $\geq 6.5\%$ on two occasions*)

Question 5:

Which statement about genetic predisposition in Type 2 diabetes is true?

1. First-degree relatives have 10-15% risk compared to general population
2. Type 2 DM follows classic Mendelian inheritance pattern
3. Genetic factors are responsible for 80-90% of diabetes risk
4. Single gene mutations are the primary cause in Asian Indians

Answer: 1 (*Strong family history increases risk, but it's multifactorial, not simple Mendelian*)

Section 2: Clinical Features & Diagnosis

Question 6:

Which of the following is NOT a classic symptom of diabetes mellitus?

1. Polyuria
2. Polydipsia
3. Polyphagia
4. Polycythemia

Answer: 4 (*Polycythemia is increased red blood cells, not a diabetes symptom*)

Question 7:

A 45-year-old man presents with recurrent infections and delayed wound healing. What diagnostic test should be performed next?

1. Lipid profile
2. Random blood glucose
3. ECG
4. Chest X-ray

Answer: 2 (*Random blood glucose is the first step when symptoms suggest hyperglycemia*)

Question 8:

According to ADA guidelines, pre-diabetes is diagnosed when fasting plasma glucose is:

1. 90-110 mg/dL
2. 100-125 mg/dL
3. 126-140 mg/dL
4. 140 mg/dL

Answer: 2 (*IFG: 100-125 mg/dL, IGT in OGTT: 140-199 mg/dL*)

Question 9:

The gold standard diagnostic test for diabetes mellitus is:

1. Fasting plasma glucose
2. Postprandial blood glucose
3. Oral glucose tolerance test (OGTT)
4. Random blood glucose

Answer: 3 (*OGTT is the gold standard as it assesses glucose metabolism under standardized conditions*)

Question 10:

Which complication screening test detects early diabetic nephropathy?

1. Fundoscopy
2. Vibration perception threshold
3. Urine albumin-creatinine ratio
4. Ankle-brachial index

Answer: 3 (*UACR detects microalbuminuria, earliest sign of diabetic nephropathy*)

Section 3: Risk Factors & Prevention

Question 11:

Which modifiable risk factor has the strongest association with Type 2 diabetes?

1. Family history
2. Obesity (BMI ≥ 30 kg/m²)
3. Age > 45 years
4. Ethnicity (Asian Indian)

Answer: 2 (*Obesity is the strongest modifiable risk factor, doubling diabetes risk*)

Question 12:

According to WHO "Best Buys" for diabetes prevention, which intervention is cost-effective at population level?

1. Individual counseling for high-risk persons
2. Corporate wellness programs
3. Taxing sugar-sweetened beverages
4. Mandatory health insurance coverage

Answer: 3 (*Fiscal policies like sugar taxation are highly cost-effective for diabetes prevention*)

Question 13:

The Diabetes Prevention Program (DPP) trial showed that intensive lifestyle intervention reduced diabetes incidence by:

1. 20-30%
2. 31-50%
3. 58%

4. 70-80%

Answer: 3 (DPP trial showed 58% reduction in diabetes incidence with lifestyle intervention over 3 years)

Question 14:

Which waist circumference cutoff is appropriate for Asian Indians according to WHO/IDF?

1. 80 cm in women, > 90 cm in men
2. 90 cm in women, > 100 cm in men
3. 80 cm in both sexes
4. 102 cm in women, > 88 cm in men

Answer: 1 (Lower cutoffs for South Asians: >80 cm women, >90 cm men)

Question 15:

Gestational diabetes mellitus (GDM) increases the lifetime risk of developing Type 2 diabetes by:

1. 2-3 fold
2. 5-7 fold
3. 10-15 fold
4. 20-25 fold

Answer: 2 (Women with GDM have 5-7 fold increased risk of developing Type 2 DM)

Section 4: Management & Treatment

Question 16:

First-line pharmacological treatment for Type 2 diabetes according to ADA/EASD 2022 consensus is:

1. Insulin glargine
2. Metformin
3. Sulfonylureas
4. DPP-4 inhibitors

Answer: 2 (Metformin remains first-line unless contraindicated)

Question 17:

Which medication class has been shown to reduce cardiovascular mortality in Type 2 diabetes?

1. Sulfonylureas
2. DPP-4 inhibitors
3. SGLT2 inhibitors
4. Thiazolidinediones

Answer: 3 (SGLT2 inhibitors like empagliflozin reduce CV mortality in EMPA-REG trial)

Question 18:

Tight glycemic control (HbA1c < 6.5%) is recommended for which group of diabetic patients?

1. Elderly patients with multiple comorbidities
2. Patients with long disease duration
3. Newly diagnosed patients without comorbidities
4. All diabetic patients regardless of age

Answer: 3 (*Individualized targets - tighter control for younger patients without CV disease*)

Question 19:

The recommended glycemic target for elderly diabetic patients (>65 years) with comorbidities is:

1. HbA1c < 6.0%
2. HbA1c < 6.5%
3. HbA1c 7.0-8.0%
4. HbA1c < 8.5%

Answer: 3 (*Relax glycemic targets in elderly with comorbidities to avoid hypoglycemia*)

Question 20:

Which insulin regimen is recommended for newly diagnosed Type 2 diabetes patients with HbA1c > 10%?

1. Single daily dose insulin
2. Split-mixed insulin twice daily
3. Multiple daily injections (basal-bolus)
4. Premixed insulin twice daily

Answer: 1 (*Start with single daily basal insulin, titrate gradually*)

Section 5: Complications & Monitoring

Question 21:

The most common chronic complication of diabetes mellitus is:

1. Diabetic retinopathy
2. Diabetic nephropathy
3. Diabetic neuropathy
4. Coronary artery disease

Answer: 3 (*Diabetic peripheral neuropathy affects 60-70% of patients*)

Question 22:

Screening for diabetic retinopathy should begin:

1. At diagnosis for Type 1 DM, after 5 years for Type 2 DM
2. At diagnosis for both Type 1 and Type 2 DM
3. After 5 years of diabetes for both types
4. After puberty for Type 1 DM, at diagnosis for Type 2 DM

Answer: 2 (*Annual screening from diagnosis for both types*)

Question 23:

What is the earliest detectable sign of diabetic nephropathy?

1. Proteinuria (albuminuria > 300 mg/day)
2. Decreased GFR (<60 mL/min/1.73m²)
3. Microalbuminuria (30-300 mg/day)
4. Creatinine clearance <30 mL/min

Answer: 3 (*Microalbuminuria is the earliest clinical sign of diabetic nephropathy*)

Question 24:

Which complication is most likely to cause hypoglycemia unawareness in diabetic patients?

1. Diabetic retinopathy
2. Diabetic nephropathy
3. Diabetic neuropathy
4. Autonomic neuropathy

Answer: 4 (*Autonomic neuropathy impairs catecholamine responses to hypoglycemia*)

Question 25:

The leading cause of ESRD (End Stage Renal Disease) in India is:

1. Chronic glomerulonephritis
2. Diabetic nephropathy
3. Hypertensive nephrosclerosis
4. Obstructive uropathy

Answer: 2 (*Diabetic nephropathy accounts for 40-50% of ESRD cases in India*)

Section 6: Community Medicine Perspective

Question 26:

Under NPCDCS 2010, annual diabetes screening is recommended for:

1. All adults above 18 years
2. All adults above 30 years
3. All adults above 25 years with risk factors
4. All adults above 35 years

Answer: 2 (NPCDCS recommends annual screening for all adults ≥ 30 years)

Question 27:

The role of ASHA (Accredited Social Health Activist) in diabetes care includes:

1. Prescribing oral hypoglycemics
2. Conducting laboratory tests
3. Promoting awareness and follow-up
4. Performing surgical interventions

Answer: 3 (ASHAs create awareness, ensure follow-up, refer patients, not prescribe/diagnose)

Question 28:

Which national program provides free oral hypoglycemics for diagnosed diabetics?

1. RNTCP (Tuberculosis Program)
2. NPCDCS (National Program for Cancer, Diabetes, CVD)
3. RMNCHA (Reproductive, Maternal, Neonatal Health)
4. NVBDCP (Vector Borne Disease Program)

Answer: 2 (NPCDCS provides free OHA and insulin through government network)

Question 29:

The WHO "Package of Essential Noncommunicable Disease Interventions" (PEN) provides primary care tools for:

1. Specialty care diabetes management
2. Community-based diabetes prevention and care
3. Tertiary level diabetes research
4. Advanced surgical interventions

Answer: 2 (PEN provides practical tools for PHC level NCD management including diabetes)

Question 30:

Which of the following is NOT part of the NP-IHP (National Programme for Improving Healthcare, 2017) diabetes component?

1. Screening and management at PHC level
2. Free drug distribution through ASHA
3. Specialty care at district hospitals
4. Mandatory insurance coverage for all diabetics

Answer: 4 (NP-IHP provides comprehensive diabetes care but insurance is through other schemes like Ayushman Bharat)

Section 7: Emergency Management

Question 31:

First-line treatment for hypoglycemia (blood glucose < 70 mg/dL) in a conscious patient is:

1. Intravenous dextrose 50%
2. Glucagon injection 1 mg
3. 15g rapid-acting carbohydrates
4. Insulin discontinuation

Answer: 3 (*Rule of 15: 15g fast-acting carbs, wait 15 minutes, recheck*)

Question 32:

Classic presentation of diabetic ketoacidosis (DKA) includes all EXCEPT:

1. Hyperglycemia (>250 mg/dL)
2. Metabolic acidosis
3. Ketonaemia/ketonuria
4. Hyponatremia

Answer: 4 (*DKA presents with hyperglycemia, acidosis, ketones; may have hyponatremia from osmotic diuresis*)

Question 33:

Priority treatment for hyperglycemic hyperosmolar state (HHS) is:

1. Potassium supplementation first
2. Insulin bolus immediately
3. Bicarbonate therapy
4. Fluid resuscitation with normal saline

Answer: 4 (*Fluid replacement is first step in HHS management to correct dehydration*)

Question 34:

A diabetic patient on insulin presents with altered sensorium and blood glucose of 45 mg/dL. What is the most appropriate immediate response?

1. Reduce next insulin dose by 25%
2. Administer subcutaneous glucagon
3. Provide 15g fast-acting carbohydrates orally
4. Arrange immediate hospitalization

Answer: 3 (*First treat hypoglycemia with oral carbs if patient is conscious; glucagon IM if unconscious*)

Question 35:

In DKA management, insulin should be initiated when serum potassium is:

1. < 3.5 mEq/L (give potassium first)
2. ≥ 3.5 mEq/L (safe to start insulin)
3. < 5.0 mEq/L (dangerous hyperkalemia)

4. ≥ 5.5 mEq/L (requires immediate treatment)

Answer: 2 (Never give insulin when $K^+ < 3.5$ mEq/L due to risk of arrhythmia; initiate only when normokalemia achieved)

Section 8: Lifestyle & Patient Education

Question 36:

According to ADA guidelines, adults with diabetes should perform at least how much moderate-intensity exercise per week?

1. 75 minutes
2. 120 minutes
3. 150 minutes
4. 300 minutes

Answer: 3 (ADA recommends 150 minutes moderate intensity + muscle strengthening 2x/week)

Question 37:

Dietary recommendation for diabetic patients regarding carbohydrate intake is:

1. Low carbohydrate diet (< 50 g/day)
2. Carbohydrate 45-65% of total calories
3. Carbohydrate 70-75% of total calories
4. No restriction, all carbohydrates equal

Answer: 2 (Carbohydrate 45-65% of total calories, focusing on complex carbohydrates and glycemic load)

Question 38:

Self-monitoring of blood glucose (SMBG) is most beneficial for:

1. All Type 2 diabetes patients on lifestyle modification only
2. Patients on insulin requiring dose adjustment
3. Patients checking once yearly for diagnostic confirmation
4. Patients with stable diabetes not requiring medication

Answer: 2 (SMBG most beneficial for insulin users, pregnant patients, hypoglycemia-prone patients)

Question 39:

The most effective strategy for permanent lifestyle change in diabetes management is:

1. Frequent clinic visits every 2 weeks
2. Technology-based monitoring (apps, trackers)
3. Family and social support systems
4. Monetary incentives for weight loss

Answer: 3 (Family support and social support systems are most effective for sustainable lifestyle changes)

Question 40:

Which patient education method shows the highest retention rate in diabetes care?

1. Didactic lectures by physicians
2. Printed pamphlets and booklets
3. One-on-one counseling sessions
4. Interactive group discussions with role-play

Answer: 4 (*Interactive methods with discussion, demonstration, and practice lead to better retention*)

Answer Key Summary

Epidemiology & Pathophysiology: 1-3, 2-4, 3-2, 4-2, 5-1 **Clinical Features & Diagnosis:** 6-4, 7-2, 8-2, 9-3, 10-3 **Risk Factors & Prevention:** 11-2, 12-3, 13-3, 14-1, 15-2 **Management & Treatment:** 16-2, 17-3, 18-3, 19-3, 20-1 **Complications & Monitoring:** 21-3, 22-2, 23-3, 24-4, 25-2 **Community Medicine:** 26-2, 27-3, 28-2, 29-2, 30-4 **Emergency Management:** 31-3, 32-4, 33-4, 34-3, 35-2 **Lifestyle & Education:** 36-3, 37-2, 38-2, 39-3, 40-4

Total Questions: 40

Distribution: 5 questions per section × 8 sections **Difficulty Level:** Mix of recall, application, and analysis questions **Reference:** ADA 2023, WHO-PEN, NCDIR, IDF Guidelines

Note: These MCQs are designed for CBME assessment, testing knowledge application rather than mere recall. Answers include brief explanations for learning reinforcement.

Created for: MBBS Community Medicine CBME Diabetes Training Module **Coordinator:** Dr. Siddalingaiah H S
Last Updated: September 2025